

ViPragSent: XLM-R-large for Vietnamese Pragmatic Sentiment Analysis

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Abstract

Pragmatic sentiment in Vietnamese text is not exhausted by polarity. Sarcasm, irony, idiomaticity, implicitness, code-switching, and mocking can change how a reader should interpret an utterance even when its surface sentiment is positive or negative. We introduce ViPragSent, an XLM-R-large encoder with six distinct binary pragmatic heads, separate three-way polarity and seven-way emotion auxiliaries, and a training-only teacher-forced rationale objective. The model uses a shared first-nonpadding-token representation and homoscedastic uncertainty weighting to couple these tasks while keeping their semantic roles separate. On a matched 2,000-example ID/gold test cohort, ViPragSent reaches 93.67 ± 0.19 macro-pragmatic F1 versus 92.93 ± 0.14 for the standard XLM-R baseline, an observed +0.73 percentage-point difference. Its mean is also highest on all six pragmatic heads and on the macro average among five complete three-seed baseline families. Follow-up analyses show that the full objective is an operating point rather than a uniform winner: Q2 reports accuracy, calibration, and cost trade-offs; Q3 reports a non-monotonic positive-label budget curve with contextual baselines; and Q4 reports target ECE 0.0386 ± 0.0069 versus Vistral 0.0345 ± 0.0041 . These results define what the implemented model contributes while keeping rationale supervision separate from inference.

1 Introduction

Sentiment analysis is often framed as a polarity problem, but pragmatic meaning can reverse or qualify the interpretation suggested by a surface-positive or surface-negative expression. A Vietnamese utterance may be sarcastic, ironic, idiomatic, implicit, code-switched, or mocking without belonging to an emotion

category. These phenomena are therefore useful prediction targets in their own right. Treating them as six emotion classes would collapse distinct communicative functions and obscure the role of emotion as an auxiliary signal. ViPragSent adopts the narrower formulation: six pragmatic phenomena are six binary tasks, while polarity and emotion are separate auxiliary tasks.

Vietnamese offers a demanding setting. PhoBERT shows the value of language-specific pretraining (Nguyen and Nguyen, 2020), whereas XLM-R provides a multilingual transfer regime across one hundred languages (Conneau et al., 2020). ViSoBERT applies the XLM-R architecture to Vietnamese social-media tasks (Nguyen et al., 2023b). These contrasting settings motivate testing a large multilingual encoder on a bundle of pragmatic distinctions rather than one sentiment label.

The challenge is also objective design. Pragmatic labels may correlate with polarity and emotion without being identical. Multi-task learning can use related signals (Caruana, 1997), while uncertainty weighting adapts relative loss scales (Kendall et al., 2018). These are empirical choices, so we report the training losses, causal rationale decoder, and classification-head inference interface explicitly.

This paper studies ViPragSent XLM-R-large through four linked questions:

- **Q1a–Q1b:** How does the primary model behave in-domain, and how much of that behavior is retained on external Vietnamese sentiment and emotion datasets?
- **Q2:** What changes when the polarity, emotion, explanation, multitask, or task-uncertainty components are removed?

- **Q3:** How does pragmatic performance change as the number of positive examples is constrained while the evaluation procedure is held fixed?
- **Q4:** How well calibrated are the resulting confidence estimates relative to comparison models under a common expected-calibration-error protocol?

The contribution is a focused analysis of one primary XLM-R-large model rather than a leaderboard claim. We contribute (i) an implemented shared encoder with separated pragmatic, polarity, emotion, and teacher-forced rationale objectives; (ii) a six-head comparison on a matched ID/gold prediction cohort; and (iii) follow-up analyses that expose accuracy, calibration, cost, and budget trade-offs. All reported means summarize three saved runs, and comparison claims concern the evaluated prediction artifacts rather than a fresh training or inference rerun.

2 Related Work

2.1 Multilingual and Vietnamese encoders

XLM-R established a strong multilingual masked-language-model baseline by scaling pretraining across languages and evaluating cross-lingual transfer (Conneau et al., 2020). PhoBERT demonstrates the complementary case for a Vietnamese-specific encoder trained on word-segmented Vietnamese data (Nguyen and Nguyen, 2020). ViSoBERT moves toward Vietnamese social-media text while retaining the XLM-R architecture (Nguyen et al., 2023b). ViPragSent uses XLM-R-large as its primary encoder so that the study can focus on the pragmatic multi-task formulation and its auxiliary objectives.

The Q1a comparison contains standard XLM-R, PhoBERT, Sailor, and Vistral prediction artifacts. ViSoBERT is discussed here as contextual prior work, not as an evaluated Q1a baseline: no ViSoBERT prediction family is included in the verified comparison table. This distinction prevents a related model family from being mistaken for a measured result.

2.2 Pragmatic and affective supervision

The target formulation combines signals that are related but not interchangeable. UIT-VSFC provides Vietnamese student feedback with sentiment-oriented labels (Nguyen et al., 2018), whereas UIT-VSMEC is a Vietnamese social-media emotion corpus (Ho et al., 2020). ViPragSent uses these label families as auxiliary signals while retaining separate pragmatic outputs.

Sarcasm detection studies show why context can matter beyond the current utterance: conversation-aware models can outperform models that read only the current turn (Ghosh et al., 2018). A complementary line of work distills teacher-generated rationales as additional supervision in a multi-task objective (Hsieh et al., 2023); earlier rationale models likewise separate prediction from explanation (Lei et al., 2016). ViPragSent brings this rationale-supervision idea to a shared XLM-R-large encoder and tests it alongside Vietnamese pragmatic heads, while keeping rationale generation outside the deployed prediction interface.

2.3 Multi-task objectives, rationale supervision, and calibration

Multi-task learning shares a representation across related objectives and can transfer useful domain information between them (Caruana, 1997). The uncertainty-weighted objective follows the motivation that task losses need not contribute equally (Kendall et al., 2018). The implemented rationale decoder supplies a teacher-forced auxiliary loss during training and is discarded at inference.

Confidence is evaluated separately from discrimination: expected calibration error measures the gap between confidence and empirical correctness across bins, which can remain large when accuracy is strong (Guo et al., 2017). Q4 is therefore a protocol-specific calibration comparison, not a universal uncertainty claim.

3 Method

3.1 Task definition

Given a Vietnamese utterance x , ViPragSent predicts six binary pragmatic labels in the

manifest order: implicit sentiment, sarcasm, irony, idiom/figurative language, code-switching, and mocking. These labels represent distinct pragmatic phenomena. The model may also predict a three-way polarity label and a seven-way emotion label. The auxiliary labels are not additional pragmatic classes.

For each pragmatic head, the metric is binary macro-F1: the arithmetic mean of the F1 values for class 0 and class 1. The macro-pragmatic F1 is the arithmetic mean of the six binary macro-F1 values. This definition is important for the head-level table because it differs from a positive-class-only F1.

3.2 XLM-R-large encoder and prediction heads

The primary model uses FacebookAI/xlm-roberta-large at revision c23d21b0620b635a76227c604d44e43a9f0ee389. The encoder returns a hidden state for each input token. For the XLM-R family, the implementation pools the first nonpadding representation: the hidden state at the first position with an active attention mask, rather than a mean over the sequence. A 0.1 dropout layer feeds task-specific linear heads: six one-logit binary heads, one three-logit polarity head, and one seven-logit emotion head. At inference, predictions come from these classification heads.

3.3 Implemented auxiliary objective

The six pragmatic heads use weighted binary cross-entropy with logits, with positive-class weights computed from the training split. Polarity and emotion use class-weighted cross-entropy. For the full model, the eight classification losses have independent learned log-variance parameters. Let $\mathcal{T}$ be the eight classification tasks, let m_t be the recorded task multiplier, and let s_t be the learned log variance. The implementation clamps the log variance before applying uncertainty weighting:

$$\begin{aligned} s_t^c &= \text{clip}(s_t, -5, 5), \\ \mathcal{L}_{\text{cls+rat}} &= \sum_{t \in \mathcal{T}} \left[\frac{1}{2} e^{-s_t^c} (m_t \ell_t) + \frac{1}{2} s_t^c \right] \\ &\quad + \beta \mathcal{L}_{\text{rat}}, \quad \beta = 0.3. \end{aligned}$$

Here ℓ_t is the task loss before the multiplier, so the multiplier is applied inside the uncertainty-

weighted classification term. The rationale term is present only for the full rationale-enabled variant.

The rationale component is a two-layer causal **TransformerDecoder**. It uses a memory projection from the encoder hidden size to a decoder hidden size of 128, four attention heads, feed-forward size 512, dropout 0.1, causal masking, and tied token input/output embeddings. During training it consumes rationale tokens with teacher forcing and contributes token cross-entropy as $\beta \mathcal{L}_{\text{rat}}$ with $\beta = 0.3$. For the target run, the manifest identifies `approved_generated_rationales_train.jsonl` with 7,998 training rows. The inspected evidence does not identify a provider for these generated targets, so we treat this as a run-level training artifact rather than evidence about the original dataset’s annotation history. The decoder is discarded from the inference path and produces no reported prediction.

3.4 Training and selection

The frozen ViPragSent package is identified in the data handoff as the SEACrowd/ViSoBERT source package and contains 11,997 rows: 7,998 training, 1,999 development, and 2,000 test examples. The split seed is 20260520. Training uses date-coded seeds 20260521, 20260522, and 20260523, reported in the paper as logical seeds 21, 22, and 23. The primary Q1a v37 configuration uses AdamW with learning rate 2×10^{-5} , weight decay 0.01, a maximum of 10 epochs, bf16 precision, physical batch size 8 with four accumulation steps (effective batch size 32), gradient clipping 1.0, cosine scheduling, and a 0.20 warmup ratio. Early stopping patience is 10. The primary loss multipliers are 1.01 for implicit sentiment, 1.01 for sarcasm, 1.05 for irony, and 1.04 for code-switching; the other task multipliers are 1.0. The selection metric is development macro-pragmatic F1.

For each binary pragmatic head, the development threshold is selected from 0.05 through 0.95 in steps of 0.01 by binary macro-F1, with ties resolved toward 0.5. Q3 applies this rule independently for each seed and budget. Q4 uses raw positive-class sigmoid probabilities and therefore does not threshold the probabil-

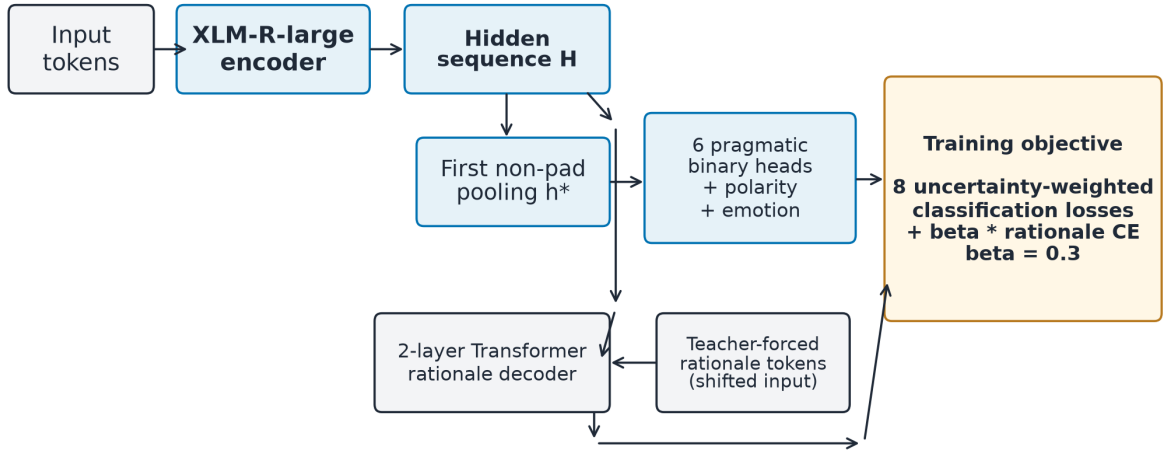


Figure 1: ViPragSent architecture and training objective. XLM-R-large produces a hidden sequence that feeds first-nonpadding pooling for six pragmatic binary heads, three-way polarity, and seven-way emotion prediction. The same hidden sequence is used as memory by a two-layer rationale decoder with teacher-forced shifted rationale tokens during training only; its rationale loss contributes to the objective and is not an inference output.

ities before computing ECE.

3.5 Provenance and comparison scope

The Q1a evidence package contains 18 saved prediction JSONL files: three ViPragSent target runs and three runs for each of five complete baseline families. Every one of the 45 target–baseline pairs has 2,000 rows with the same sample-ID set and the same six pragmatic gold-label tuples when aligned by sample ID. The Q1a scores were deterministically recomputed from those saved rows using the metric above. This supports a direct score comparison for the evaluated ID/gold cohort. It does not claim identical upstream input text, identical training, a significance test, or an independent rerun.

4 Experimental Setup

4.1 Separate experimental cohorts

Q1a, Q2, Q3, and Q4 are related but distinct artifact cohorts. The distinction is material when interpreting the full-model values.

The Q2 full value is not a rerun of Q1a v37: the cohorts differ in scheduler, warmup ratio, and early-stopping patience. Q2 is therefore interpreted within its own follow-up cohort, not as an explanation of the difference between Q1a’s 93.67 and Q2’s 92.8.

4.2 Data and external evaluation

Q1a evaluates the 2,000-example pragmatic test cohort. Q1b reports retention on UIT-VSFC, UIT-VSMEC, and AIVIVN. The external aggregate is called ordinary F1 in this paper: it is the unweighted mean of the three external macro-F1 values. It is not ordinal F1 and does not encode an ordinal relation among labels. External datasets have different domains and label semantics, so their scores are not pooled with the primary macro-pragmatic F1.

4.3 Baselines and reporting conventions

ViPragSent XLM-R-large is the only primary model. Comparison records include standard XLM-R, PhoBERT single-task and fine-tuned variants, Sailor, and Vistral where complete three-seed summaries are available. Vistral ViPragSent variants and incomplete GPT records are outside the primary Q1a comparison. ViSoBERT is contextual prior work, not an evaluated baseline in this study.

The available baseline manifests describe family-specific recipes rather than one recipe shared with the target. The standard XLM-R baseline uses AdamW at 2×10^{-5} , bf16, physical batch size 8 with four accumulation steps, linear scheduling with 0.10 warmup,

Cohort	Run family and purpose	Key protocol
Q1a	Optimized v37 ViPragSent and five complete baseline families	Cosine scheduler, warmup 0.20, patience 10, development macro-pragmatic-F1 selection; target task multipliers as above
Q2	XLM-R follow-up full model and five ablations	Linear scheduler, warmup 0.10, patience 2; named auxiliary and uncertainty variants
Q3	XLM-R follow-up nested positive-label budgets	Linear scheduler, warmup 0.10, patience 2; development sarcasm binary macro-F1 selection
Q4	Calibration extraction	Same-seed extraction from each Q1a v37 full checkpoint; no new training

Experimental cohorts and protocols; quantitative entries elsewhere summarize three saved runs where reported ($n = 3$, mean $\pm$ sample SD).

and a ten-epoch maximum; its selection metric is development macro-pragmatic F1. The PhoBERT fine-tune uses AdamW at 2×10^{-5} , bf16, physical batch size 32, linear scheduling with 0.10 warmup, and the same ten-epoch maximum. The PhoBERT single-task record is a bundle of six separately trained component artifacts and exposes no single aggregate optimizer recipe. Sailor-7B and Vistral-7B-Chat are causal-7B pragmatic-SFT records (Dou et al., 2024; Nguyen et al., 2023a) using paged AdamW 8-bit, learning rate 10^{-4} , bf16, physical batch size 2 with eight accumulation steps, cosine scheduling, 0.05 warmup, and a three-epoch maximum; their reported inference source is the classification heads. These records are described for provenance and interpretation only: the ViPragSent v37 cosine schedule, warmup, patience, and task multipliers are not claimed for the baselines.

Unless otherwise stated, a value is the arithmetic mean over three saved runs and the uncertainty is the sample standard deviation

with $n = 3$. Q1a and Q3 report F1 in percentage points. Q2 reports macro-pragmatic F1, ECE multiplied by 10^3 , and relative cost. Q4 reports ECE on its native $[0,1]$ scale. Lower ECE is better. No significance test is applied to the three-seed summaries.

5 Results

5.1 Q1a: in-domain pragmatic performance

Table 1 gives the main result at the level of all six pragmatic phenomena and their macro average. Values are mean $\pm$ sample SD in percentage points over seeds 21, 22, and 23. The target is highest in mean on every head and on the macro average among the five complete-seed baseline families. The strongest baseline is standard XLM-R for every row.

Against the strongest baseline, the observed mean gaps are +0.51 pp for implicit sentiment, +1.24 pp for sarcasm, +0.50 pp for irony, +0.50 pp for idiom/figurative language, +0.57 pp for code-switching, and +1.07 pp for mocking. The two largest observed head gaps are therefore sarcasm and mocking, but these are descriptive differences across saved predictions, not significance claims. The macro difference is $93.665838 - 92.934251 = +0.731587$ percentage points, reported as +0.73 pp.

5.2 Q1b: external retention

Table 2 shows mixed transfer rather than uniformly preserved performance. The ViPragSent XLM-R record is higher than Sailor on UIT-VSFC and ordinary F1, lower on UIT-VSMEC and AIVIVN, and differs in variability. Vistral is included as comparison context.

All entries are mean $\pm$ sample SD over three saved runs. The external datasets differ in domain and target semantics, so the table is a retention profile rather than a single generalization score. The larger spread on some target rows also cautions against interpreting a single mean as stable transfer.

5.3 Q2: auxiliary-task and uncertainty ablations

Table 3 reports the separate XLM-R follow-up cohort. Within this cohort, the full configuration is not the highest reported macro-

Phenomenon	ViPragSent XLM-R-large	XLM-R base- line	PhoBERT single	PhoBERT fine-tune	Sailor	Vistral
Implicit sentiment	92.67 $\pm$ 0.24	92.16 $\pm$ 0.13	88.00 $\pm$ 1.29	85.76 $\pm$ 0.72	85.23 $\pm$ 0.11	87.68 $\pm$ 0.42
Sarcasm	89.36 $\pm$ 0.90	88.12 $\pm$ 0.43	70.87 $\pm$ 1.34	72.64 $\pm$ 0.50	72.24 $\pm$ 0.50	75.40 $\pm$ 1.06
Irony	98.43 $\pm$ 0.20	97.93 $\pm$ 0.10	97.13 $\pm$ 1.06	97.45 $\pm$ 0.19	96.40 $\pm$ 0.55	97.02 $\pm$ 0.01
Idiom/ figurative	93.35 $\pm$ 0.66	92.85 $\pm$ 0.68	91.21 $\pm$ 0.78	86.63 $\pm$ 0.66	91.17 $\pm$ 0.41	91.52 $\pm$ 0.35
Code-switching	94.84 $\pm$ 0.56	94.27 $\pm$ 0.04	91.08 $\pm$ 1.51	91.73 $\pm$ 0.53	90.48 $\pm$ 0.87	93.22 $\pm$ 0.45
Mocking	93.35 $\pm$ 0.73	92.27 $\pm$ 0.41	79.16 $\pm$ 2.99	78.53 $\pm$ 1.01	77.82 $\pm$ 0.08	81.16 $\pm$ 1.16
Macro-pragmatic F1	93.67 $\pm$ 0.19	92.93 $\pm$ 0.14	86.24 $\pm$ 0.49	85.46 $\pm$ 0.36	85.56 $\pm$ 0.10	87.67 $\pm$ 0.06

Table 1: Q1a in-domain pragmatic performance on the 2,000-example ID/gold test split. Entries are mean $\pm$ sample SD over $n = 3$ logical runs (21, 22, 23), in percentage points of binary macro-F1 per phenomenon and macro-pragmatic F1.

Model record	UIT-VSFC	UIT-VSMEC	AIVIVN	Ordinary F1
Sailor	34.4 $\pm$ 3.2	40.7 $\pm$ 0.7	48.7 $\pm$ 1.4	41.3 $\pm$ 1.2
ViPragSent	37.9 $\pm$ 2.8	40.0 $\pm$ 0.7	47.9 $\pm$ 2.2	42.0 $\pm$ 1.8
XLM-R-large				
Vistral	30.7 $\pm$ 6.4	39.4 $\pm$ 1.5	49.0 $\pm$ 0.8	39.7 $\pm$ 2.5

Table 2: External retention on the UIT-VSFC, UIT-VSMEC, and AIVIVN test cohorts. Entries are mean $\pm$ sample SD over $n = 3$ saved runs; columns report macro-F1 in percentage points and their unweighted ordinary-F1 aggregate.

pragmatic F1: removing task-uncertainty weighting gives 92.9 versus 92.8 for the full configuration. The full configuration has lower reported ECE than the no-uncertainty variant, while removing the explanation auxiliary reduces relative cost substantially and changes F1 and ECE. Removing the multitask bundle causes a much larger F1 reduction. These are trade-offs within Q2, not evidence that one component is universally beneficial.

The metric artifacts expose `polarity_dev_ece`; Table 3 combines test-split macro-pragmatic F1 with this saved development-split three-way ECE on the 10^3 scale. It is not Q4’s six-head calibration metric or a new top-label recomputation. Because Q2 is a separate cohort, its scientific comparison is the pattern across named variants, not an explanation of Q1a’s absolute score.

5.4 Q3: pragmatic label-budget behavior

Table 4 reports the XLM-R follow-up budget profile together with contextual PhoBERT and Vistral baseline curves. Values are mean $\pm$ sample SD over three runs. Thresholds are tuned on the fixed development split; the displayed metrics are evaluated on the fixed test split.

The target exceeds the available contextual

XLM-R variant	Macro- pragmatic F1	Saved polarity ECE $\times 10^3$	Relative cost
Full	92.8 $\pm$ 0.6	78.3 $\pm$ 5.9	1.00
follow-up			
No emotion auxiliary	91.7 $\pm$ 0.7	61.8 $\pm$ 27.0	0.65
No explanation auxiliary	92.3 $\pm$ 1.3	68.4 $\pm$ 21.2	0.20
No multitask bundle	74.7 $\pm$ 2.3	115.3 $\pm$ 70.1	1.10
No polarity auxiliary	92.8 $\pm$ 0.2	N/A	0.81
No task- uncertainty weighting	92.9 $\pm$ 0.3	81.6 $\pm$ 10.8	0.82

Table 3: Q2 follow-up ablations. Macro-pragmatic F1 is evaluated on the Q2 test split; the saved `polarity_dev_ece` field is a development-split three-way polarity ECE, shown multiplied by 10^3 ; relative cost is normalized to the full Q2 XLM-R run. Entries are mean $\pm$ sample SD over $n = 3$ runs.

baseline rows at displayed budgets, but the curves are non-monotonic: macro-pragmatic F1 rises from 91.1 at 32 positives to 92.3 at 512 and returns to 91.1 at full budget. Missing 512/full PhoBERT cells limit the baseline curve. This is a sample-efficiency diagnostic, not evidence that less data is generally preferable; the supplied plot visualizes the same cohort.

Positive examples	ViPragSent sarcasm / macro	PhoBERT sarcasm / macro	Vistral sarcasm / macro
32	84.2 $\pm$ 1.6 / 91.1 $\pm$ 1.3	69.7 $\pm$ 1.5 / 84.2 $\pm$ 0.7	70.5 $\pm$ 2.4 / 86.2 $\pm$ 0.2
64	83.7 $\pm$ 1.3 / 91.2 $\pm$ 0.8	70.8 $\pm$ 2.8 / 77.7 $\pm$ 11.8	72.5 $\pm$ 0.9 / 86.7 $\pm$ 0.5
128	85.2 $\pm$ 0.7 / 91.6 $\pm$ 0.2	71.8 $\pm$ 2.2 / 84.0 $\pm$ 1.2	71.3 $\pm$ 0.6 / 86.2 $\pm$ 0.4
256	85.3 $\pm$ 2.6 / 91.6 $\pm$ 1.2	73.4 $\pm$ 1.2 / 84.5 $\pm$ 0.9	64.0 $\pm$ 5.1 / 70.3 $\pm$ 14.2
512	87.2 $\pm$ 1.2 / 92.3 $\pm$ 0.7	–	73.9 $\pm$ 1.2 / 87.2 $\pm$ 0.3
Full	85.0 $\pm$ 2.3 / 91.1 $\pm$ 1.3	–	75.3 $\pm$ 0.8 / 87.4 $\pm$ 0.2

Table 4: Q3 pragmatic label-budget behavior on the fixed test split. Entries are mean $\pm$ sample SD over $n = 3$ logical runs; sarcasm and macro columns are F1 in percentage points, and – denotes an unavailable baseline cell.

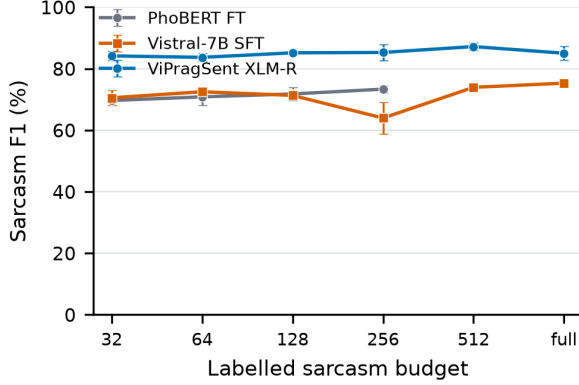


Figure 2: Q3 sarcasm F1 across pragmatic-label budgets. Curves show artifact-derived means and sample SD over three logical runs ($n = 3$; run labels 21, 22, and 23, corresponding to remote artifact seeds 20260521, 20260522, and 20260523). The target is shown with PhoBERT fine-tune and Vistral-7B SFT for context.

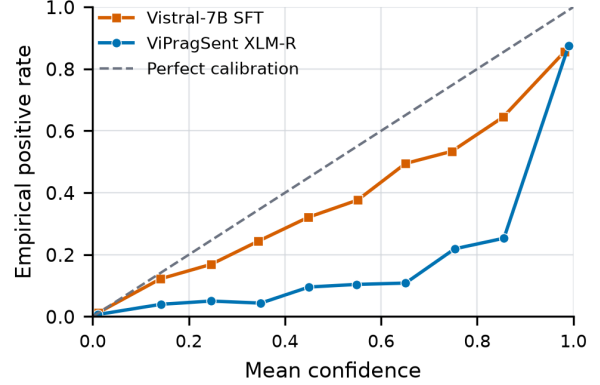


Figure 3: Q4 reliability diagrams for the calibration comparison. Curves aggregate the supplied per-label reliability bins over three logical runs ($n = 3$; run labels 21, 22, and 23, corresponding to remote artifact seeds 20260521, 20260522, and 20260523). The dashed line denotes perfect calibration; the figure is descriptive and does not imply target dominance.

5.5 Q4: calibration

Under the Q4 extraction protocol, the ViPragSent XLM-R target has ECE 0.0386 ± 0.0069 , while the Vistral comparison record has ECE 0.0345 ± 0.0041 . Lower ECE is better, so the comparison does not support target calibration dominance. Q4 extracts raw positive sigmoid probabilities from same-seed Q1a v37 checkpoints without retraining; ECE depends on its split, ten bins, and probability source.

6 Analysis and Discussion

6.1 What the primary result establishes

Q1a establishes a direct score comparison for the saved matched cohort: all 45 pairs share 2,000 sample IDs and six pragmatic gold tuples, and macro-F1 was deterministically recomputed. ViPragSent records 93.67 versus 92.93 for standard XLM-R (+0.73 pp), leads

on all six heads, and has the largest observed gaps on sarcasm (+1.24 pp) and mocking (+1.07 pp). It has the highest recorded mean among the five complete baseline families, but the result remains descriptive because recipes differ.

6.2 Auxiliary supervision is an operating-point choice

Q2 shows that auxiliary supervision is an operating-point choice. The full variant has 92.8 macro-pragmatic F1, while removing uncertainty weighting is slightly higher; the other variants trade F1, ECE, and cost in different directions, with the no-multitask bundle showing a large F1 reduction. These three-run comparisons do not isolate a causal mechanism. Q2 is a separate cohort, not an explanation of Q1a’s 93.67.

The rationale component is auxiliary teacher-forced token supervision during train-

ing; classification heads remain the deployed interface. The results therefore support a representation-learning interpretation, not faithful natural-language explanations at inference.

6.3 Transfer and calibration expose different weaknesses

Q1b and Q4 keep in-domain performance from carrying the argument: external retention is mixed, and Vistral has lower ECE under the supplied Q4 protocol. Discrimination, transfer, and calibration answer different questions, so deployment selection should match the operating point rather than a single macro-F1 value.

6.4 Label budgets and variance

Q3 shows why positive-label budgets need run variation. The best displayed target mean occurs at 512 positives, while the full budget does not dominate; missing PhoBERT cells limit the contextual comparison. Threshold selection, class balance, and optimization variance may contribute, but the artifacts do not isolate them. Budget studies therefore need a fixed negative pool, explicit thresholds, contextual baselines, and uncertainty summaries.

7 Limitations

The paper analyzes saved artifacts rather than an independent training or inference rerun; deterministic recomputation verifies the Q1a arithmetic but not fresh-run reproducibility. Q1a is scoped to the matched ID/gold prediction cohort, while Q2 and Q3 are separate follow-up run families and Q4 extracts calibration from Q1a v37 checkpoints. External retention is finite and heterogeneous across UIT-VSFC, UIT-VSMEC, and AIVIVN, and ECE depends on its specified split, binning, and probability source. The operational pragmatic labels do not exhaust discourse or speaker intent, and the evidence does not establish group-conditional fairness, annotation history, consent, licensing, or institutional-review details.

8 Ethics and Responsible Use

The data handoff identifies a frozen ViPragSent package sourced through

SEACrowd/ViSoBERT and records raw-text preservation, split sizes, and label fields. The current project record does not provide sufficient evidence for this paper to assert consent procedures, institutional review or exemption, compensation, licensing terms, or de-identification beyond those manifest fields. We make no such claims.

Vietnamese text with pragmatic and affective signals may contain offensive, personal, or socially sensitive language. Classification errors may mischaracterize speakers or amplify stereotypes. The model is therefore an analysis aid requiring human review, not an autonomous adjudicator of a person’s intent, emotion, credibility, or identity. Aggregate macro-F1 does not establish equal performance across social groups, domains, or writing styles. The manuscript does not reproduce individual example text, and the system is not intended for high-impact decisions or inference of protected attributes.

9 Conclusion

ViPragSent is an XLM-R-large multi-task model with separate pragmatic, polarity, and emotion heads and a training-only rationale objective. On the matched Q1a ID/gold cohort it records 93.67 ± 0.19 macro-pragmatic F1 versus 92.93 ± 0.14 for standard XLM-R (+0.73 pp), with the highest saved mean on all six heads among five complete baseline families. Q1b is mixed, Q2 exposes accuracy/calibration/cost trade-offs, Q3 is non-monotonic with contextual baselines, and Q4 shows no blanket calibration advantage. The primary result is descriptive and cohort-specific, but it supports the implemented model as a focused account of Vietnamese pragmatic prediction.

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A Reproducibility record

The following record is part of the manuscript rather than a future action list. All paths are relative to the paper workspace unless a source path is explicitly identified.

Primary backbone FacebookAI/xlm-roberta-large, revision c23d21b0620b635a76227c604d44e43a9f0ee389

Model interface First nonpadding-token pooling; six binary pragmatic heads, three-way polarity head, seven-way emotion head; classification heads are the inference source

Rationale implementation Two-layer causal TransformerDecoder with memory projection; hidden 128, four heads, feed-forward 512, dropout 0.1; tied token embeddings; teacher forcing; beta 0.3; target maximum length 160; decoder discarded at inference

Rationale target provenance Target manifest artifact approved_generated_rationales_train.jsonl; 7,998 training rows; provider unspecified in the inspected evidence; not treated as original annotation-history evidence

Dataset package SEACrowd/ViSoBERT source package; 11,997 rows; train/dev/test = 7,998/1,999/2,000; split seed 20260520; local processed

685	fingerprint 7C39BEEBC462D1F9076F5DF1	evidence/q1a_numeric_claim_ledger	731
686	565E924BD884263D5503F23398BD83A6	r.json	732
687	BF4205EB		
688	Q1a target artifact provenance Remote	Method evidence evidence/method_	733
689	target dataset fingerprint B906C090	evidence.json, evidence/evidenc	734
690	400BAE115C9C5E3C35E32FA410AC519A	e_claim_ledger.json	735
691	E09209EBA741F198087C24F9		
692	Q1a baseline artifact provenance	Protocol lineage evidence/protoco	736
693	Remote baseline dataset fingerprint	l_lineage.json; records distinct Q1a	737
694	A13573E38550ABD55D7F63E983C602BE	v37, Q2, Q3, and Q4 run families and Q4	738
695	A13C2765A1FFECA285F718478532AF0D	source-checkpoint lineage	739
696	Training seeds Date-coded seeds 20260521,	Prediction rows raw_fairness/*.jsonl;	740
697	20260522, 20260523; logical labels 21, 22,	18 files, 2,000 rows per file; deterministic	741
698	23	metric recomputation matches recorded	742
699		values	743
700	Primary Q1a optimizer AdamW, learning		
701	rate 2e-5, weight decay 0.01, bf16, phys-	The Q1a comparison covers 45 target-	744
702	ical batch 8, accumulation 4, effective	baseline pairs across the five complete baseline	745
703	batch 32, maximum 10 epochs, patience	families. Each pair has the same sample-ID set	746
704	10, gradient clipping 1.0	and six-label gold tuples after alignment by	747
705		ID. The comparison is descriptive over saved	748
706	Primary Q1a schedule Cosine, warmup ra-	predictions. No independent training or infer-	749
707	tio 0.20, development macro-pragmatic-	ence rerun is part of this record. The three	750
708	F1 selection; task multipliers implicit	fingerprints above are retained as separate lo-	751
709	1.01, sarcasm 1.01, irony 1.05, code-	cal and remote provenance identifiers; the di-	752
710	switching 1.04, remaining tasks 1.0	rect score comparison is scoped to the verified	753
711		same-ID, same-gold cohort rather than treated	754
712	Follow-up Q2/Q3 schedule AdamW,	as a fingerprint-equality test.	755
713	learning rate 2e-5, bf16, physical batch		
714	8, accumulation 4, effective batch 32,	B Table-specific protocol and	756
715	maximum 10 epochs, linear schedule,	missingness	757
716	warmup ratio 0.10, patience 2		
717	Threshold rule Candidate thresholds 0.05–	Q1a uses the optimized v37 target and the	758
718	0.95, step 0.01; development binary	complete three-seed baseline families listed in	759
719	macro-F1; ties toward 0.5	Table 1. The target and standard XLM-R	760
720		macro means are 93.665838 and 92.934251,	761
721	Q3 mask rule Nested positive subsets of 32,	with sample SDs 0.187121 and 0.144790 per-	762
722	64, 128, 256, 512, and full; fixed neg-	centage points. Their difference is 0.731587	763
723	ative pool; positive weights recomputed	percentage points. The per-head means and	764
724	per budget; development threshold per	SDs in Table 1 were recomputed from the same	765
725	seed/budget	saved JSONL prediction rows.	766
726		Q1b uses the external retention records and	767
727	Q4 rule Six pragmatic heads; raw positive	reports ordinary F1 as the unweighted mean	768
728	sigmoid probabilities; ten equal-width	of the three external macro-F1 values. Q2	769
729	bins; no temperature scaling; test split;	uses the six follow-up variants: full, no emo-	770
730	mean and sample SD across seeds; no new	tion auxiliary, no polarity auxiliary, no ratio-	771
	training	nale, no multitask, and no uncertainty weight-	772
		ing. The selected Q2 metric artifacts expose	773
	Q1a evidence evidence/q1a_fairness_co	the field polarity_dev_ece; Table 3 reports	774
	mparison.json, evidence/q1a_fai	macro-pragmatic F1 on the Q2 test split and	775
	rness_run_table.csv, evidence	the saved three-way polarity ECE from the de-	776
	/q1a_verified_table_inputs.json,	velopment split on the 10 ³ scale. The paper	777
		does not expand that field into an unverified	778

top-label recomputation. Relative cost is normalized to the full XLM-R Q2 run using the recorded mean GPU-hour values. Q3 reports the six nested target budgets and the available contextual baseline rows; missing baseline budget cells are not imputed. Q4 extracts raw probabilities from the same-seed Q1a v37 full checkpoints.

Incomplete GPT records, ViSoBERT prediction records, and excluded ViPragSent Vistral variants are not inserted into any table. Missing ECE for the Q2 no-polarity row is reported as N/A. These exclusions define the comparison scope and do not support claims about systems without verified rows.

The structure-reference PDF supplied organization only. Its historical metrics, training protocol, annotation or ethics assertions, hardware claims, and generative-decoder description are not evidence for this manuscript.